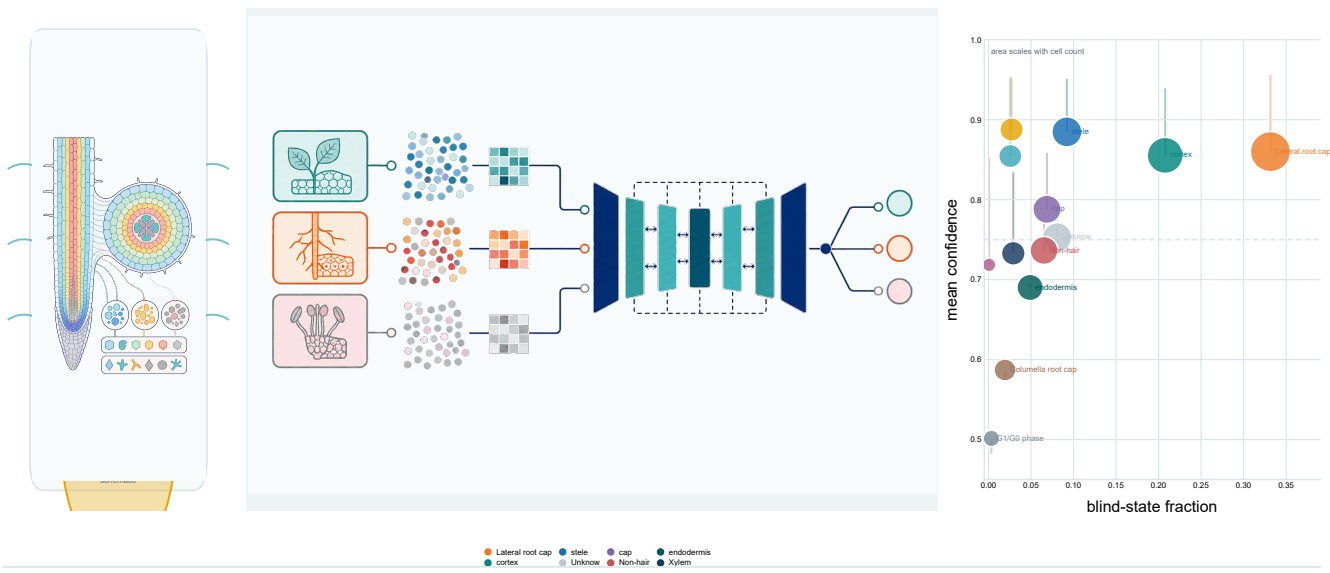


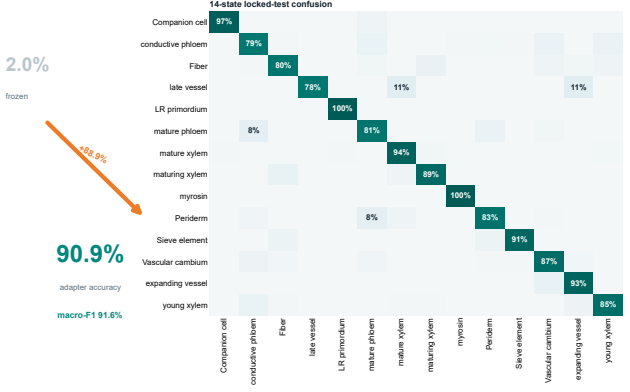
a Blind root-state atlas and confidence geometry



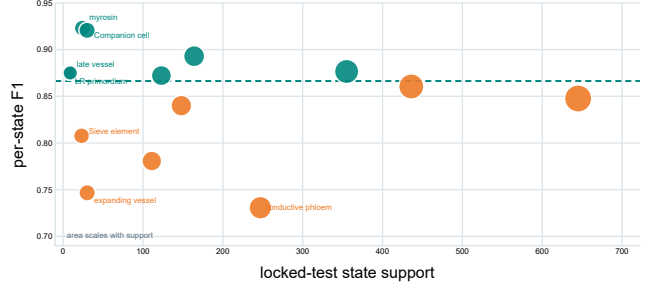
b Predeclared marker-to-identity coherence



c Locked secondary-root recovery



d State support and F1



e Validation-only model selection

